Discusión 2.0

Saturday, October 17, 2026 9:00 AM

$$V(r) = -\frac{\kappa}{r} + \frac{\lambda}{r^2} \qquad \kappa>0,\ \lambda>0$$

1) Construir $V_{ef}(r)$ e identifiqu [. . .]

$$V_{ef} = V(r) + \frac{L^2}{2\mu r^2}$$

$$V_{ef} = -\frac{\kappa}{r} + \frac{\lambda}{r^2} + \frac{L^2}{2\mu r^2}$$

$$V_{ef} = -\frac{\kappa}{r} + \frac{2\lambda\mu + L^2}{2\mu r^2}$$

$$\frac{\lambda}{r^2} + \frac{L^2}{2\mu r^2}$$

$$\frac{\lambda 2\mu}{2\mu r^2} + \frac{L^2}{2\mu r^2}$$

$$\frac{2\lambda\mu + L^2}{2\mu r^2}$$

Una órbita circular de radio r_0 es un punto de equilibrio del potencial

$r = r_0 \Rightarrow \dot{r} = 0 \quad \ddot{r} = 0$

$$\left.\frac{dV_{ef}}{dr}\right|_{r_0} = 0$$

$$\frac{dV_{ef}}{dr} = \frac{\kappa}{r^2} - \frac{4\lambda\mu + 2L^2}{2\mu r^3}$$

$$\frac{2\lambda\mu + L^2}{\mu r^3}$$

$$\frac{\kappa}{r_0^2} - \frac{4\lambda\mu + 2L^2}{2\mu r_0^3} = 0$$

$$\frac{\kappa}{r_0^{\cancel{2}}} = \frac{4\lambda\mu + 2L^2}{2\mu r_0^{\cancel{3}}}$$

$$\kappa r_0 = \frac{4\lambda\mu + 2L^2}{2\mu}$$

$$r_0 = \frac{4\lambda\mu + 2L^2}{2\kappa\mu}$$

$$r_0 = \frac{2\mu\lambda + L^2}{\kappa\mu} \qquad \kappa r_0 = \frac{2\mu\lambda + L^2}{\mu} \qquad \kappa r_0 = 2\lambda + \frac{L^2}{\mu}$$

$$u'' + u = -\frac{m}{L^2u^2} F(1/u)$$

$$u = 1/r \qquad u'' = \frac{d^2u}{d\theta^2}$$

$$V(r) = -\frac{\kappa}{r} + \frac{\lambda}{r^2}$$

$$\frac{dV}{dr} = \frac{\kappa}{r^2} - \frac{2\lambda}{r^3}$$

$$-\frac{dV}{dr} = -\frac{\kappa}{r^2} + \frac{2\lambda}{r^3} = f(r)$$

$$f(1/u) = -\kappa u^2 + 2\lambda u^3$$

$$u'' + u = -\frac{m}{L^2u^2}(-\kappa u^2 + 2\lambda u^3)$$

$$u'' + u = -\frac{m}{L^2\cancel{u^2}}(\cancel{u^2}(-\kappa + 2\lambda u)$$

$$u'' + u = -\frac{m}{L^2}(2\lambda u - \kappa)$$

$$u'' + u + \frac{2m\lambda u}{L^2} = \frac{m\kappa}{L^2}$$

$$u'' + u\left(1 + \frac{2m\lambda}{L^2}\right) = \frac{m\kappa}{L^2}$$

$$u'' + \gamma^2 u = \frac{m\kappa}{L^2}$$

$$u(\theta) = \frac{m\kappa}{L^2+\gamma^2} + A\cos(\gamma(\theta-\theta_0))$$

$$r(\theta) = \frac{1}{\frac{m\kappa}{L^2+2m\lambda} + A\cos(\gamma(\theta-\theta_0))}$$

Determinar si es estable

$$\left.\frac{d^2V_{ef}}{dr^2}\right|_{r=r_0}$$

$$\frac{d^2V_{ef}}{dr^2} = -\frac{2\kappa}{r^3} + \frac{6\lambda\mu + 3L^2}{\mu r^4}$$

$$\left.\frac{d^2V_{ef}}{dr^2}\right|_{r=r_0} = \frac{1}{r_0^4}\left(\frac{6\lambda\mu + 3L^2}{\mu} - 2\kappa r_0\right)$$

$$\frac{6\lambda\mu + 3L^2}{\mu} - 2\left(2\lambda + \frac{L^2}{\mu}\right)$$

$$6\lambda + \frac{3L^2}{\mu} - 4\lambda - \frac{2L^2}{\mu}$$

$$2\lambda + \frac{L^2}{\mu}$$

$$= \frac{1}{r_0^4}\left(2\lambda + \frac{L^2}{\mu}\right)$$

$$\frac{2\mu\lambda}{\mu} + \frac{L^2}{\mu} = \frac{2\mu\lambda + L^2}{\mu}$$

$$= \frac{1}{r_0^4}\left(\frac{2\mu\lambda + L^2}{\mu}\right)$$

Para que sea estable

$$2\mu\lambda + L^2 > 0$$

$$L^2 = \mu\kappa r_0 - 2\mu\lambda$$

$$0 = \mu\kappa r_0 - 2\mu\lambda$$

$$2\cancel{\mu}\lambda = \cancel{\mu}\kappa r_0$$

$$\lambda = \frac{\kappa r_0}{2}$$

a menos que L^2 sea cero entonces

$$\lambda < \frac{\kappa r_0}{2}$$

Pero el termino de λ no desestabilita una órbita circular.

3. Calcule $\Delta\theta$ y $\delta\varphi$ usando $g(r) = \lambda/r^2$

$$\delta\varphi = \frac{\partial}{\partial L}\left(\frac{2\mu}{L}\int_0^{\pi} r^2 g(r)\, d\theta\right)$$

$$\frac{2\mu}{L}\int_0^{\pi} \cancel{r^2}\cdot\frac{\lambda}{\cancel{r^2}}\, d\theta = \frac{2\mu\lambda}{L}\int_0^{\pi} d\theta = \frac{2\mu\lambda}{L}\cdot\pi$$

$$\delta\varphi = \frac{\partial}{\partial L}\left(\frac{2\pi\mu\lambda}{L}\right) = -\frac{2\pi\mu\lambda}{L^2}$$

$$\Delta\theta = 2\pi + \delta\varphi = 2\pi\left(1 - \frac{\mu\lambda}{L^2}\right)$$

4. Determine el sentido de la precesion y justifiquelo fisicamente.

$\delta\varphi > 0$ gira en el sentido de la particula

$\delta\varphi < 0$ en sentido contrario

$$\delta\varphi = -\frac{2\pi\mu\lambda}{L^2}$$

$\lambda > 0$ gira en sentido de la particula

$\lambda < 0$ sentido contrario

Para $\lambda > 0$ gira en sentido contrario

Si $\lambda < 0$ gira en sentido de la particula